



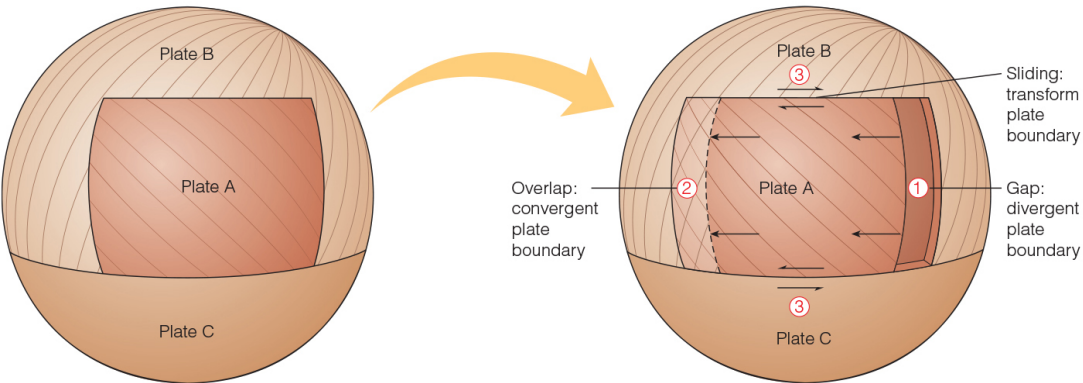
From lecture session 4: We saw the structure of the Earth and understood how the crust and lithosphere can move continents around. Remaining question: how does plate tectonics connect to creation and evolution of oceans and the shape of the ocean floor?

## Lecture Session 5

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1. Plates divergence and the formation of ocean basins;
2. Plates convergence and the recycling process of the crust;
3. Other consequences of tectonic plates and more scientific evidence.

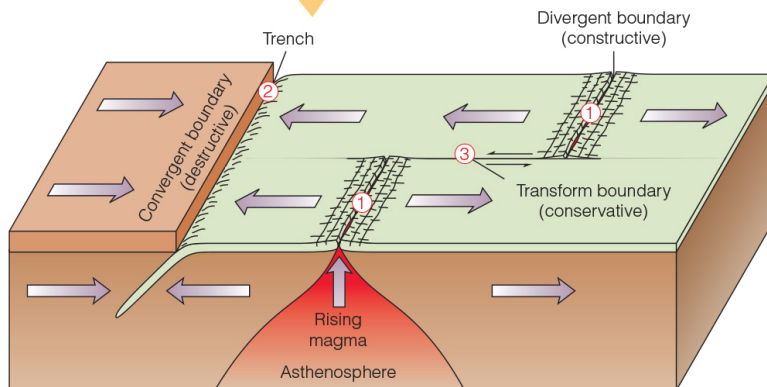
# Before we start, we need to understand the types of tectonic plate boundaries (transform, convergent, and divergent boundaries)



**a** Imagine a sphere composed of three tectonic plates.

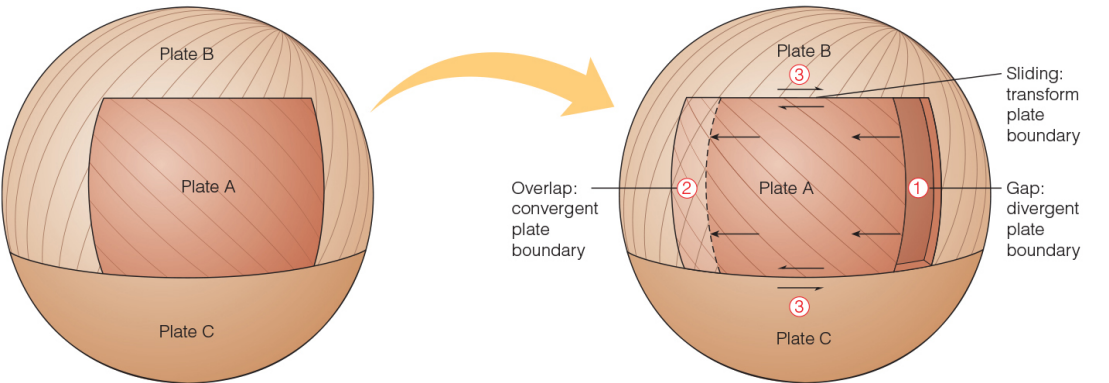
**b** Plate boundaries in action. As Plate A moves to the left (west), a gap forms behind it ①, and an overlap with Plate B forms in front ②. Sliding occurs along the top and bottom sides ③. The margins of Plate A experience the three types of interactions: At ①, extension characteristic of divergent boundaries; at ②, compression characteristic of convergent boundaries; and at ③ the shear characteristic of transform plate boundaries.

**Figure 3.17** Plate boundaries in action.



**c** Plate boundaries on the surface of a sphere. The divergent ①, convergent ②, and transform ③ margins match those of the margins of Plate A in (b).

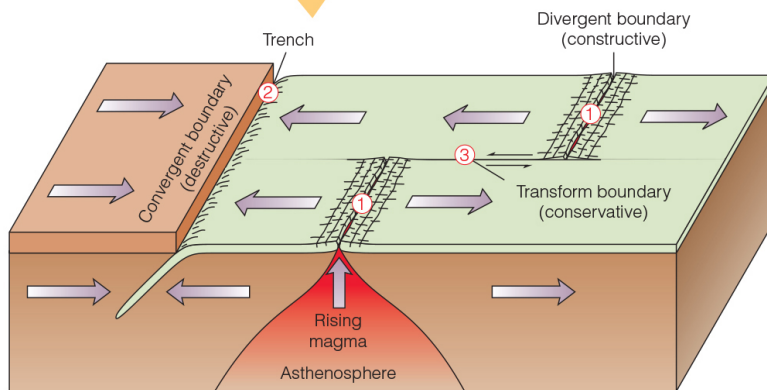
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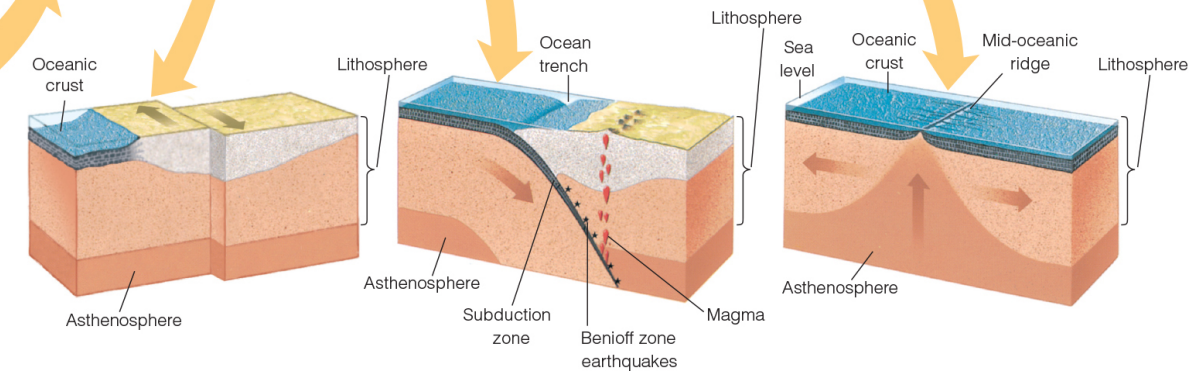
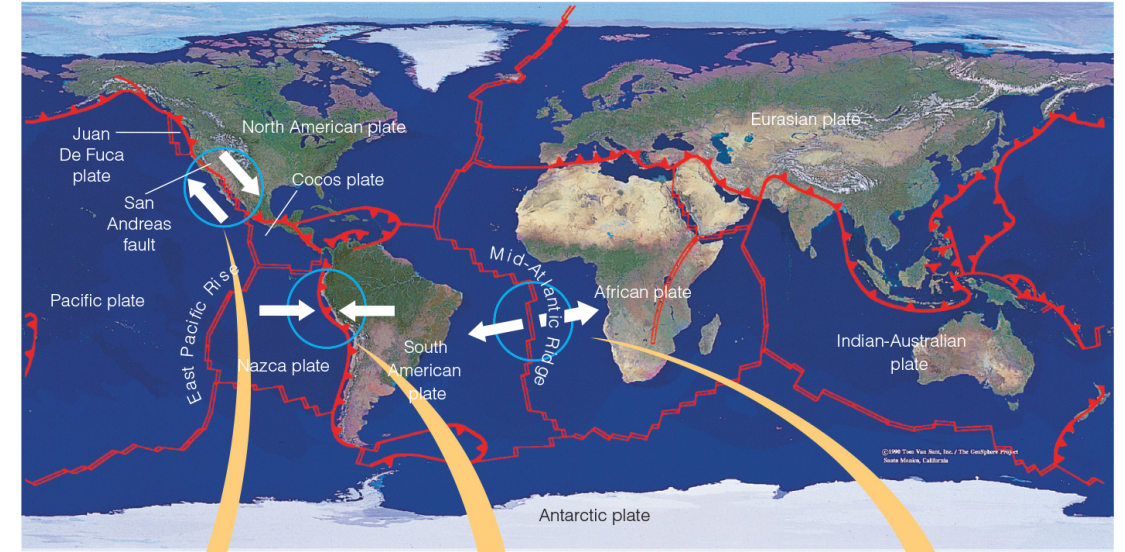
a Imagine a sphere composed of three tectonic plates.

b Plate boundaries in action. As Plate A moves to the left (west), a gap forms behind it **1**, and an overlap with Plate B forms in front **2**. Sliding occurs along the top and bottom sides **3**. The margins of Plate A experience the three types of interactions: At **1**, extension characteristic of divergent boundaries; at **2**, compression characteristic of convergent boundaries; and at **3** the shear characteristic of transform plate boundaries.

Figure 3.17 Plate boundaries in action.



c Plate boundaries on the surface of a sphere. The divergent **1**, convergent **2**, and transform **3** margins match those of the margins of Plate A in (b).

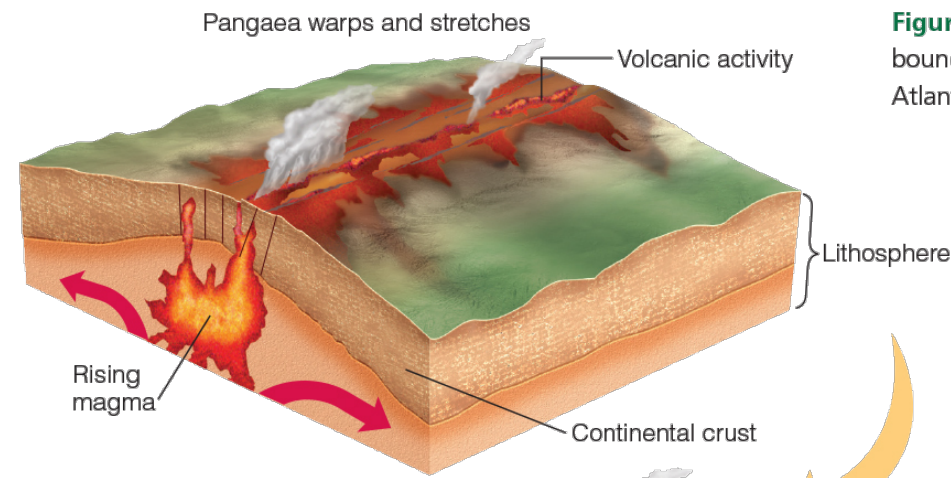


Legend:

- TRANSFORM BOUNDARY** (dashed line)
- CONVERGENT BOUNDARY** (solid line with triangles)
- DIVERGENT BOUNDARY** (solid line with arrows)

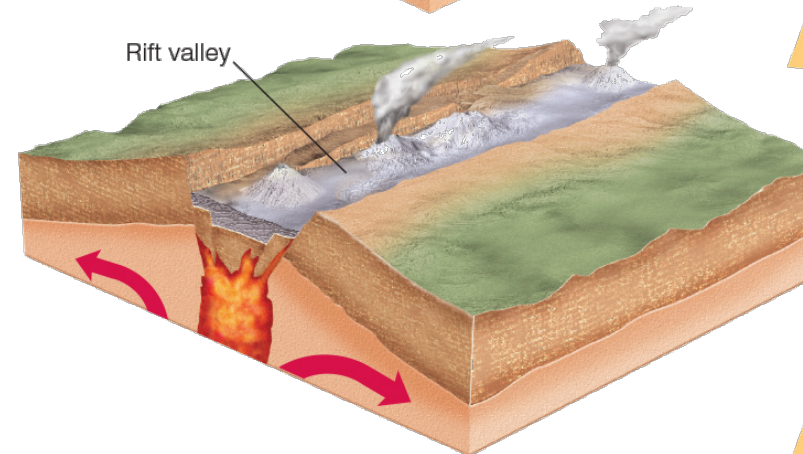
d Lateral movement at *transform* boundaries causes shear, compression at *convergent* boundaries produces buckling and shortening, and extension of *divergent* boundaries causes splitting and rifting.

# Divergent Plates and the formation of ocean basins

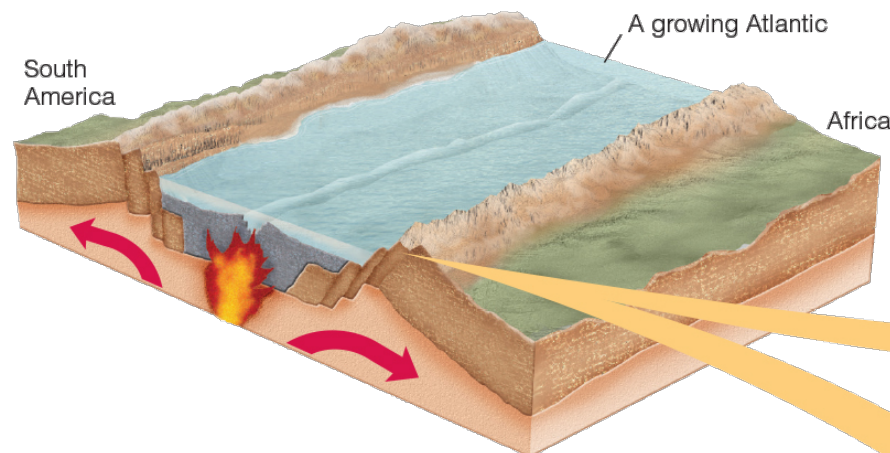


**Figure 3.18** A model for the formation of a new plate boundary: the breakup of Pangaea and the formation of the Atlantic.

**a** As the lithosphere began to crack, a rift formed beneath the continent, and molten iron and magnesium-enriched magma originating from the asthenosphere began to rise to form a new basaltic ocean floor.



**b** As the rift continued to open, the two new continents were separated by a growing ocean basin. Volcanoes and earthquakes occur along the active rift area, which is the mid-ocean (Mid-Atlantic) ridge. The East African Rift Valley, though not yet submerged, currently resembles this stage (see Figure 3.19).



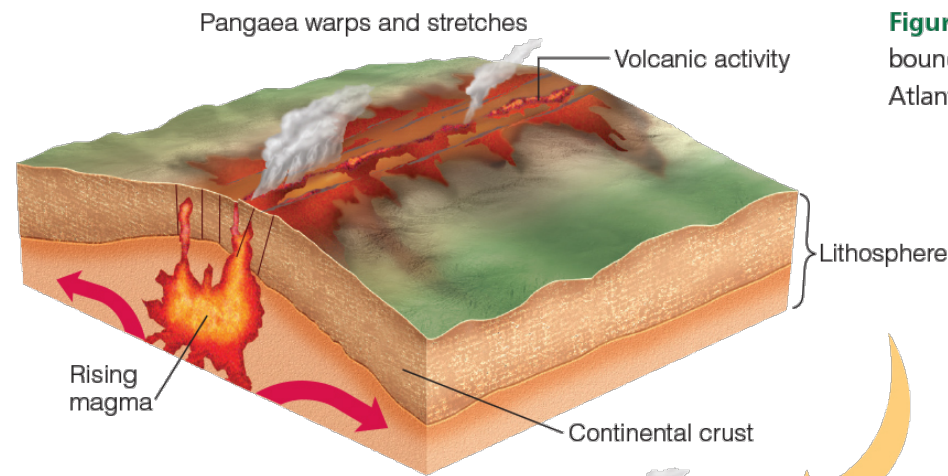
**c** A new ocean basin forms beneath a new ocean.

Jump to Figure 4.24 to see another example of the saw-tooth configuration of a mid-ocean ridge.

# Divergent Plates and the formation of ocean basins

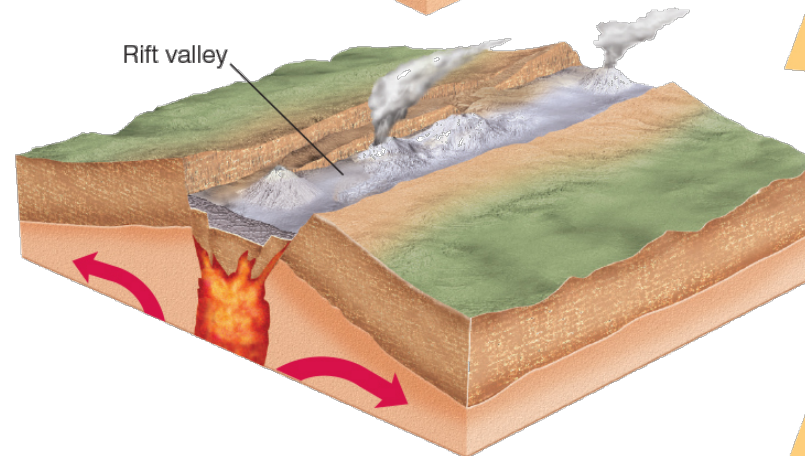
1. Separating plates due to creation of new crust;
2. Water vapor condensation (rain) and continental runoff (rivers) fill up the new formed oceanic basin;
3. During the growing phase, volcanic activity create the **Mid-Oceanic Ridges** along the divergent boundary.

Atlantic Ocean: 200 millions of years old

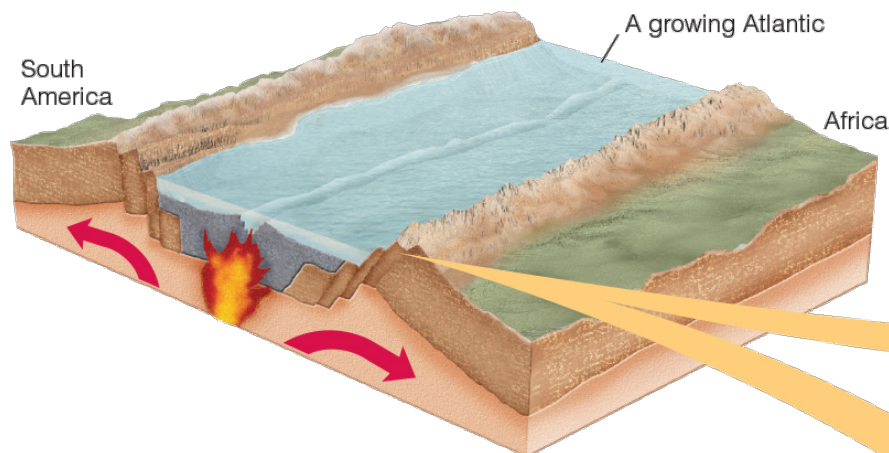


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# Plates convergence and the recycling process of the crust

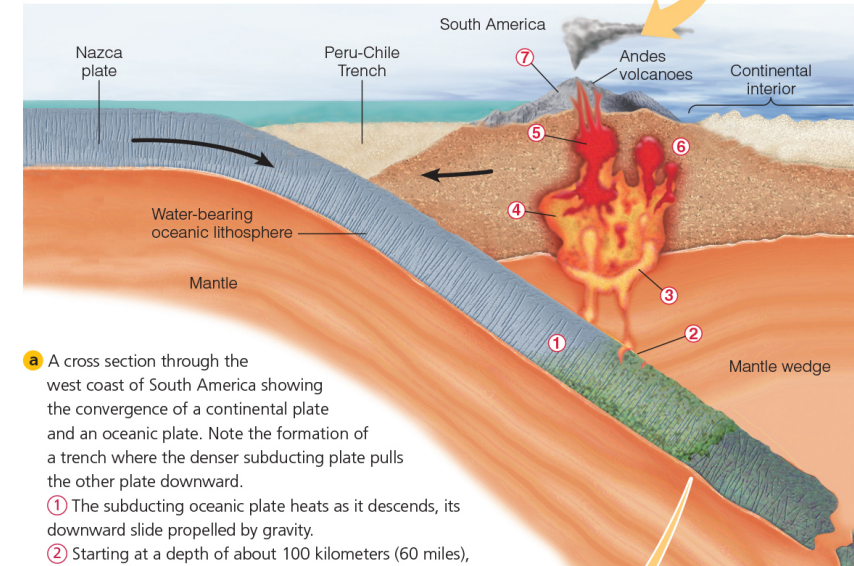
1. Denser subducted plate “dives” into the upper mantle where it will be melted and incorporated to the mantle;
2. Upper plate will deform and rise, creating **mountain ridges and volcanos**;
3. Oceanic plates subducting under continental plates are not the only example. On Earth, there are oceanic plates subduction under other oceanic plates, and continental plates subducting under continental plates.

What determine which plate will subduct?

**Density! Denser plate will always subduct.**

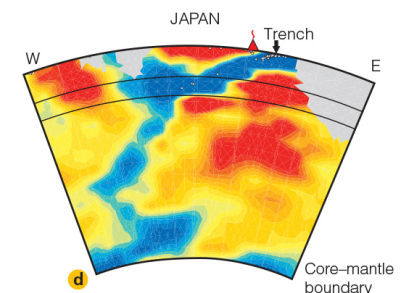
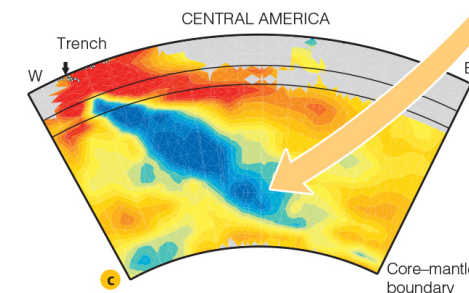
Figure 3.22 Subduction.

**b** An Andean volcano in full blast in 2006. The 5,000-meter (16,400-feet) high volcano Tungurahua, located in Ecuador, becomes active roughly every 90 years.



- a** A cross section through the west coast of South America showing the convergence of a continental plate and an oceanic plate. Note the formation of a trench where the denser subducting plate pulls the other plate downward.
- ① The subducting oceanic plate heats as it descends, its downward slide propelled by gravity.
  - ② Starting at a depth of about 100 kilometers (60 miles), heat drives water and other volatile components from the subducted plate and its sediments into the overlying mantle, lowering its melting point.
  - ③ This causes part of the overlying mantle to melt. Masses of the melted material rise and “underplate” the continental crust in places.

④ Heat from the rising material melts the continental crust and ⑤ mixes with it. Some of this mixture solidifies in place ⑥, but some can rise to the surface and power Andean volcanoes ⑦.



**c** and **d** Vertical slices through Earth's mantle beneath Central America and Japan showing the distribution of warmer (red) and colder (blue) material. The configuration of colder material suggests that the subducting slabs beneath both areas have penetrated to the core-mantle boundary, which is at a depth of about 2,900 kilometers (1,800 miles).

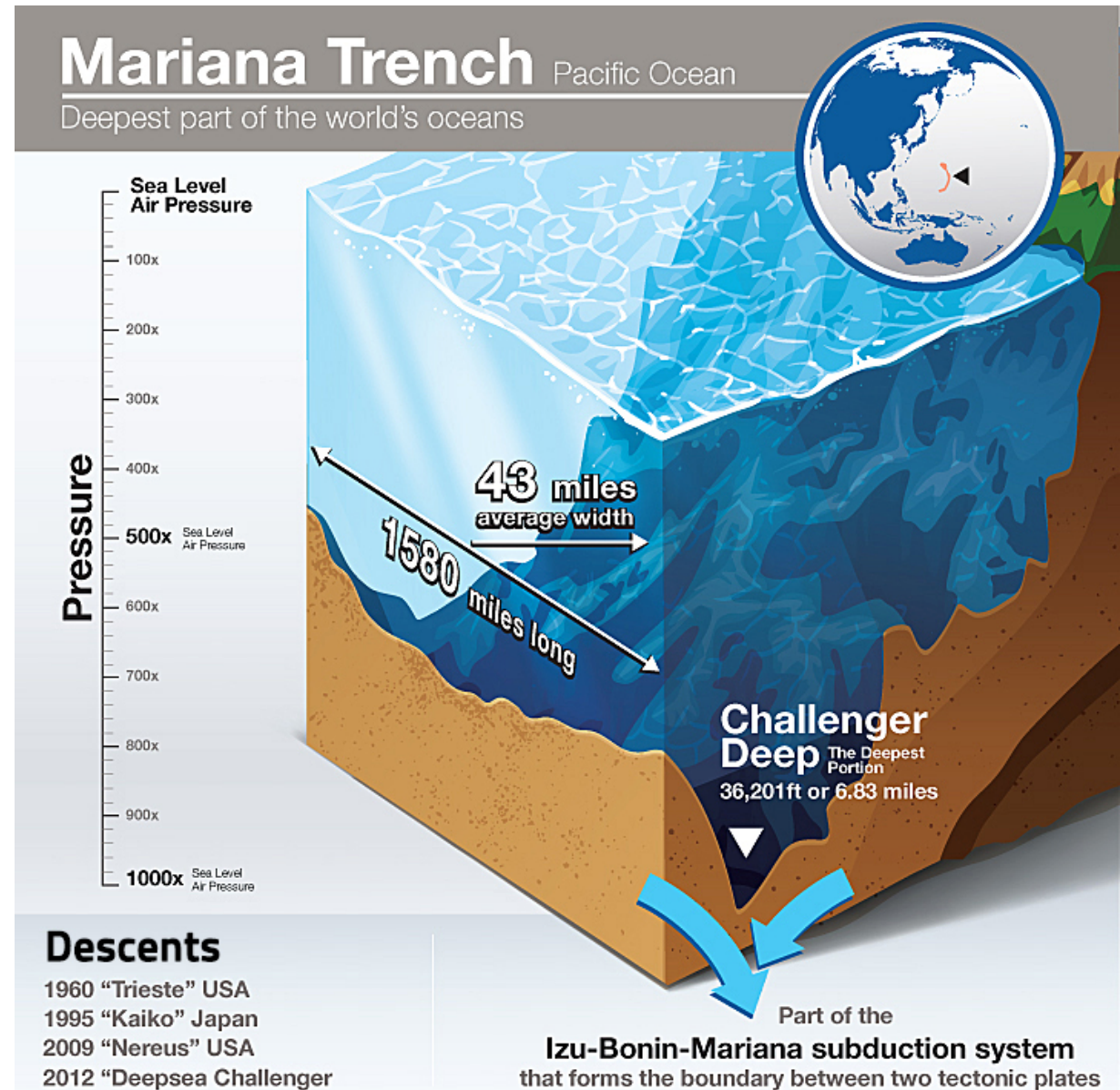
# Ocean-Continent and Ocean-Ocean plates convergence

Along convergent boundaries involving at least one oceanic plate, plates movement tend to produce oceanic trenches. These depressions represent the deepest regions of the Earth.

Deepest point of the Earth is the Challenger Deep (12,000 m deep) in the Mariana's Trench. It results from collision of two oceanic tectonic plates.

## Don't forget about the Transform Boundaries

Finally, along transform boundaries (result from lateral shear of plates) there is no destruction or creation of crust.

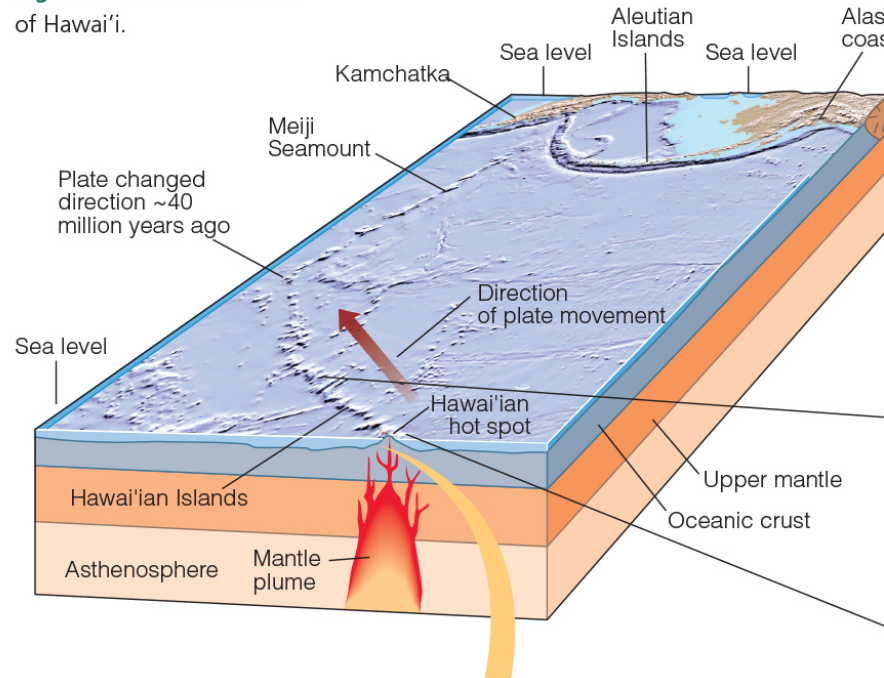


# Other consequences of moving plates

For reasons that are still unknown, there are areas of the Mantle called Mantle plumes. Such plumes extend from near the Core to the Asthenosphere, are extremely hot, are fixed, and are able to fracture/melt the overlying crust.

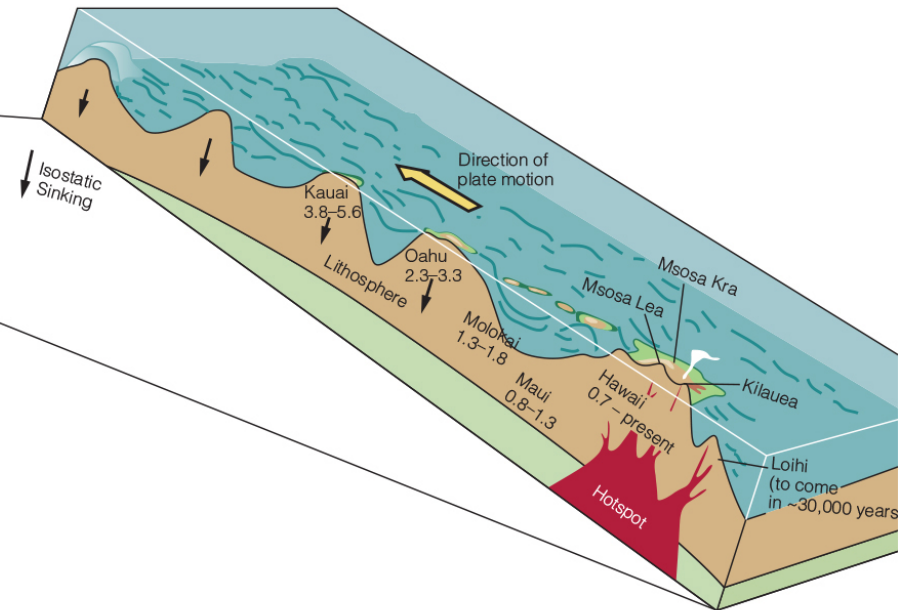
## Hot Spots and the Hawaiian Islands

**Figure 3.33** The formation of Hawai'i.



**a** Formation of a volcanic island chain as an oceanic plate moves over a stationary mantle plume and hot spot. The age of the islands increases toward the upper left. New islands will continue to form over the hot spot. In this example, showing the formation of the Hawai'ian Islands, Loihi is such a newly forming island. (All numbers given in millions of years.)

Source: NOAA National Geophysical Data Center



Once it reaches the crust's surface, the spilled magma can form islands and volcanos. As the overlying plate drifts, the volcanic active region also moves with it.

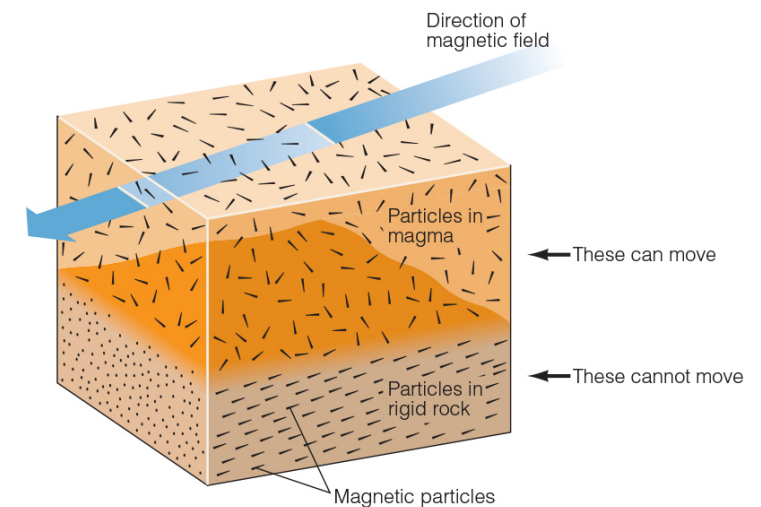
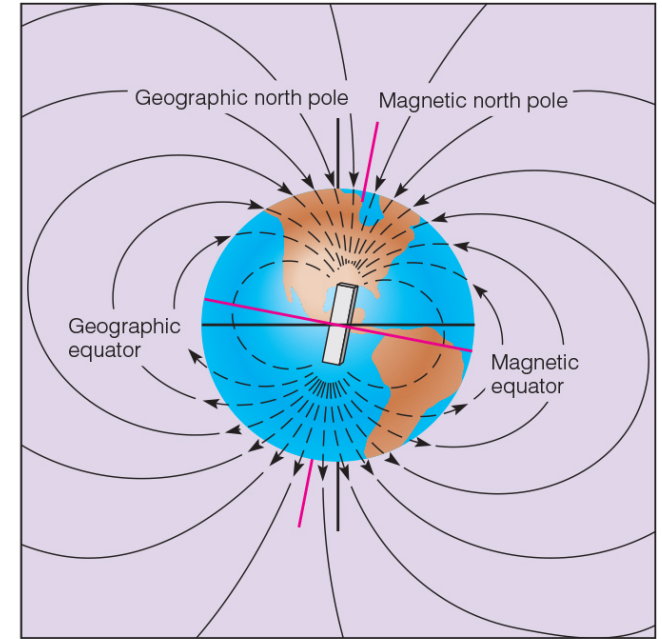


# More scientific evidence for tectonic plates: The paleomagnetism exercise

## Paleomagnetism measurements background:

The Earth's liquid outer core generates the Earth's magnetic field that protects us and our satellites from Solar storms. As time passes, the orientation of the magnetic field changes, and from time to time it can completely reverse. This means that at certain times of our geological history, compasses would point towards the South Pole. The frequency of such reversals is not fixed.

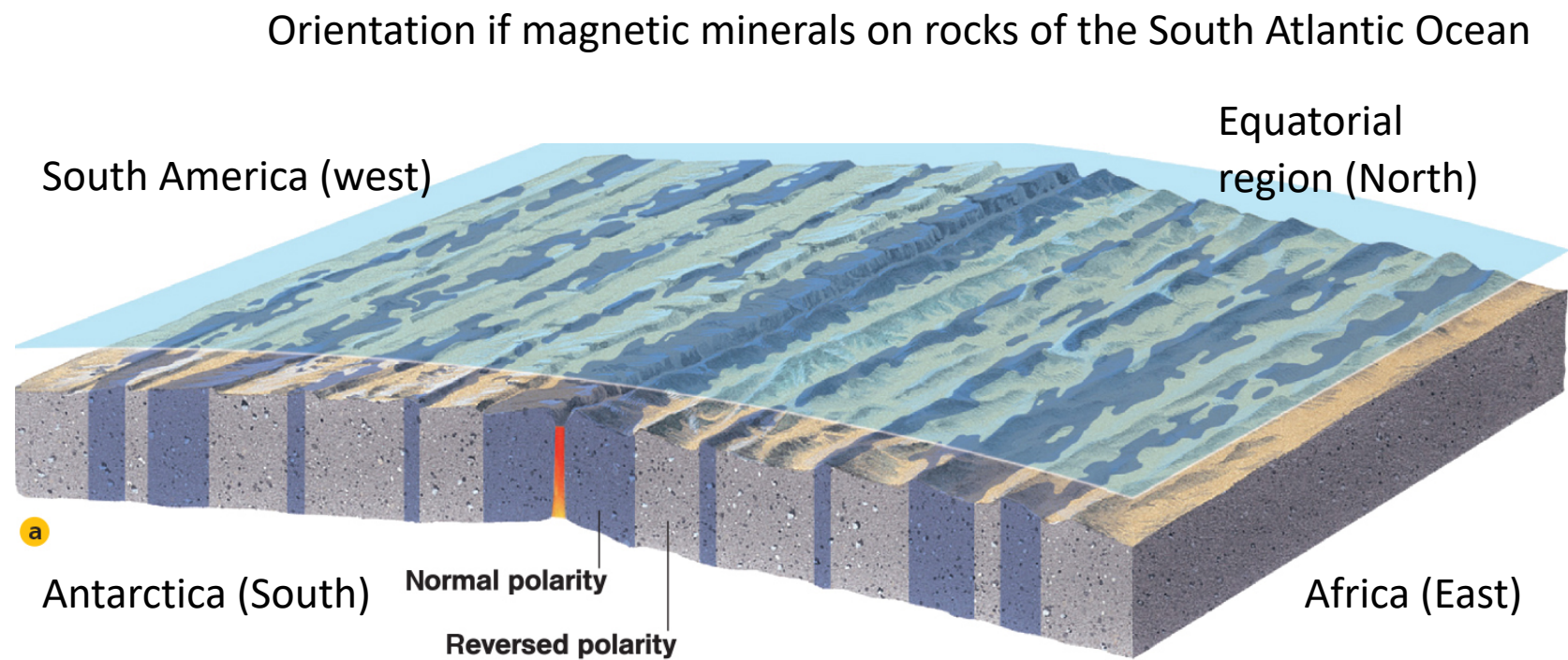
As the magma rises from the Mantle towards the crust, particles of magnetic minerals can form. As Magma reaches the surface it cools and forms rocks. Because the minerals are magnetic, they orient themselves according to the Earth's magnetic field, preserving the orientation of the field at that point in time.



## In groups of 3

You received magnetometer measurements of the orientation of the magnetic minerals from the bottom of the South Atlantic Ocean and they show a pattern as shown in the figure.

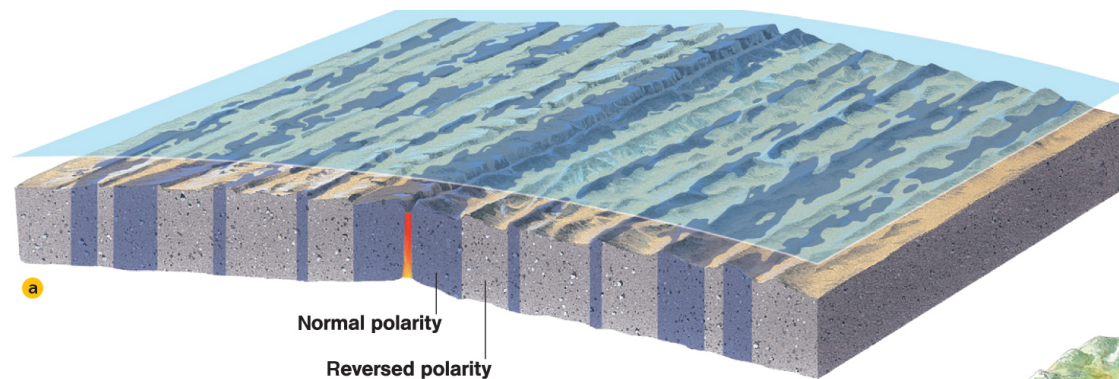
**Magnetometer:** measures the magnetic field of near rocks



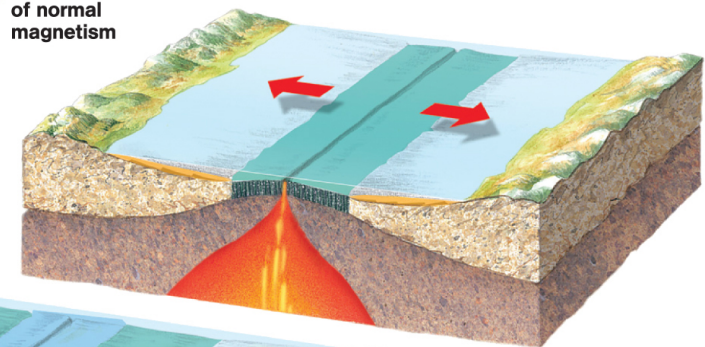
### Based on your knowledge of the Earth's magnetic field and Plate Tectonics

1. Explain the polarity (or orientation) of magnetic minerals from Atlantic rocks. Why do they change from East to West?
2. Why the polarity bands (white and blue bands) are not the same size? What are the factors contributing to the different distances between bands?
3. How would you verify if the Atlantic grows at the same rate in both sides of the Mid-Atlantic Ridge?

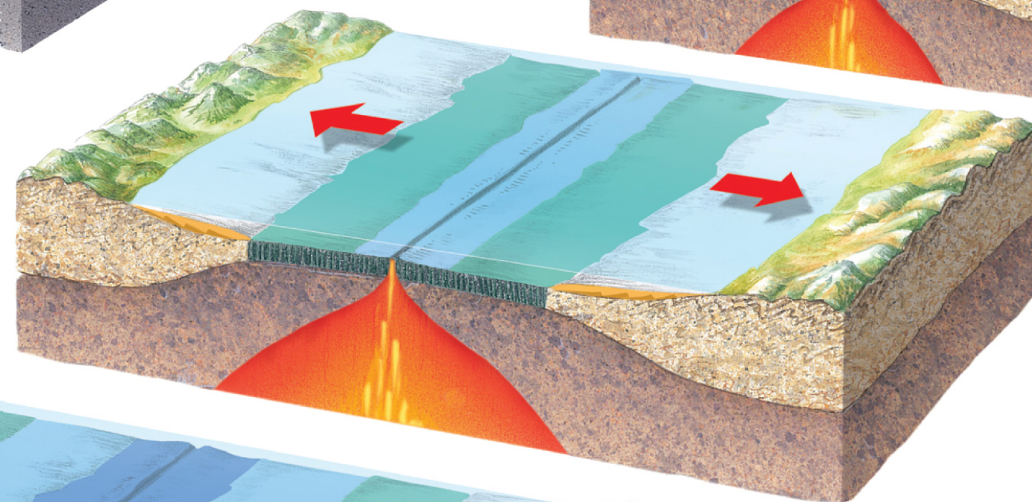
- a When scientists conducted a magnetic survey of a spreading center, the Mid-Atlantic Ridge, they found bands of weaker and stronger magnetic fields locked in the rocks.
- b-d The molten rocks forming at the spreading center take on the polarity of the planet when they are cooling and then move slowly in both directions from the center. When Earth's magnetic field reverses, the polarity of newly formed rocks changes, creating symmetrical bands of opposite polarity.



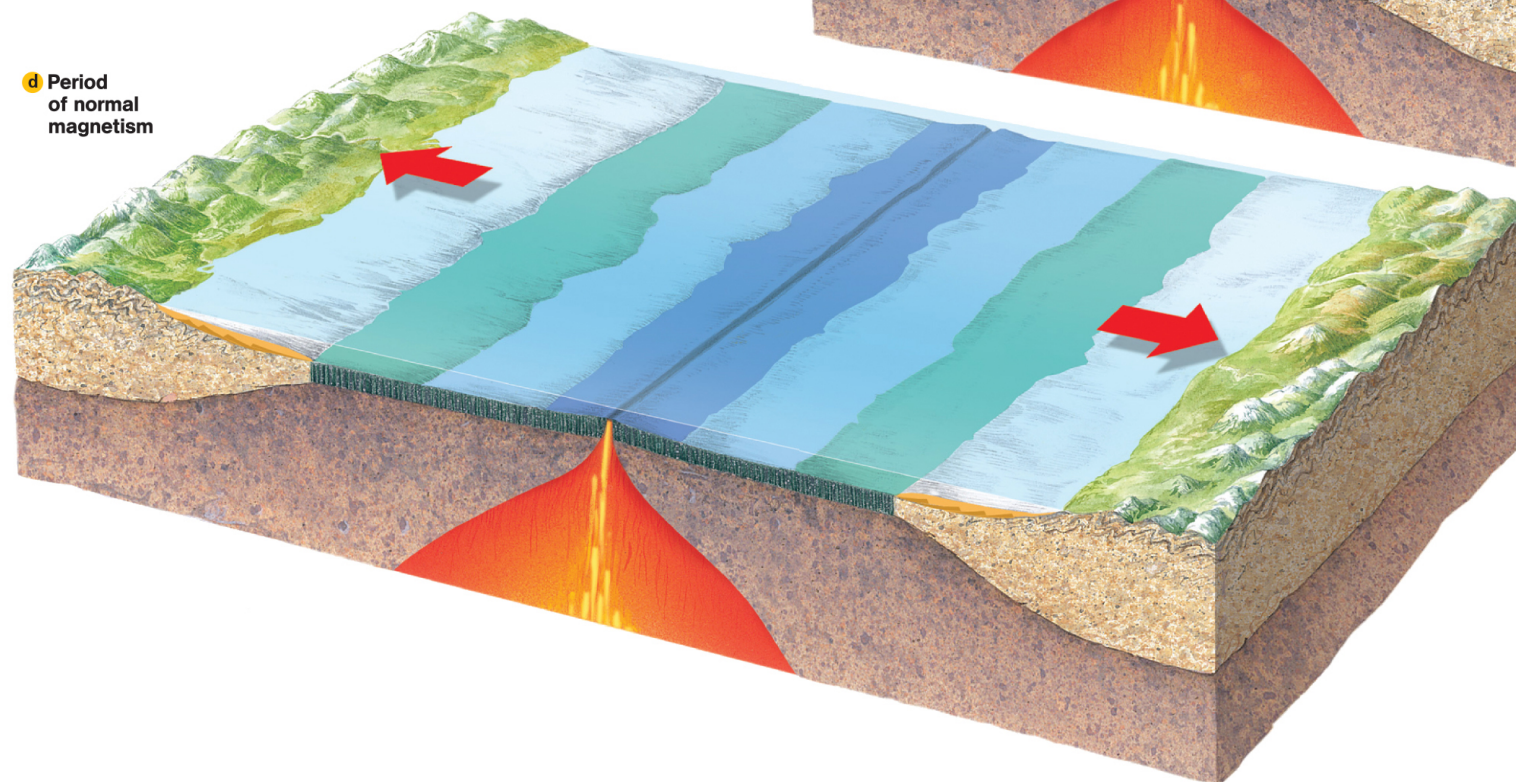
b Period of normal magnetism



c Period of reverse magnetism



d Period of normal magnetism



New bands of oceanic crust alternate in polarity as they are formed.



Normal polarity



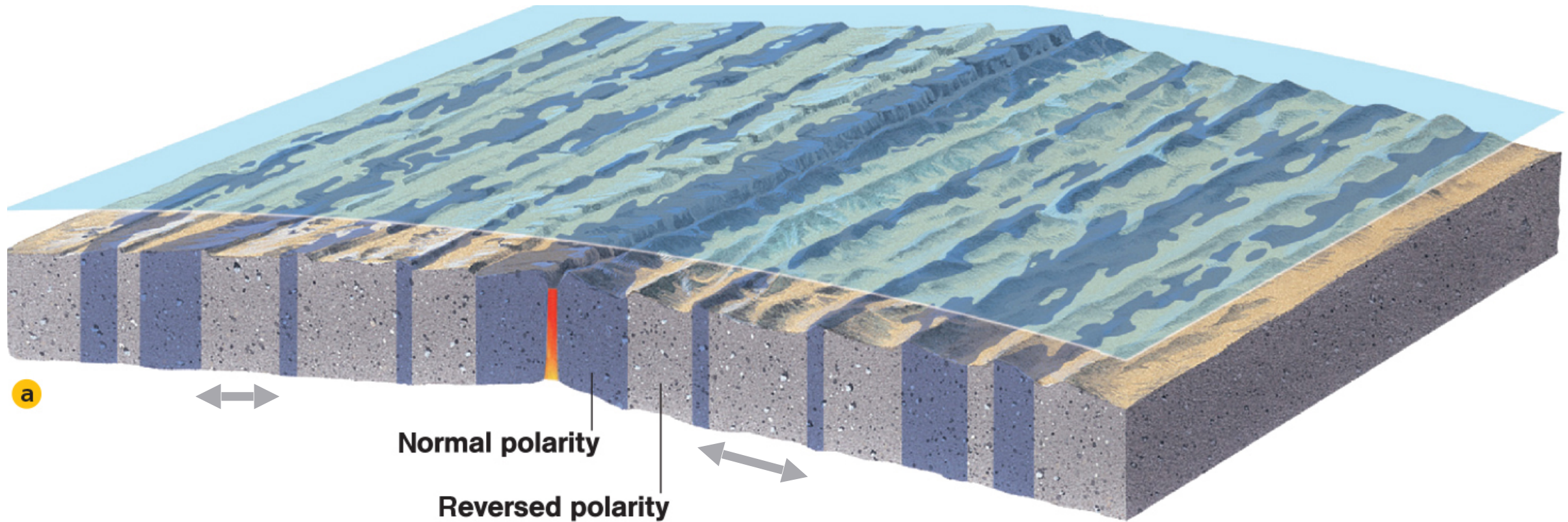
Reversed polarity



Normal polarity

1. The divergence of the South American and African plates explain the pattern. As the Atlantic opens, the newly formed crust (rocks) has the polarity of the formation time.

The polarity does not change North to South because the Atlantic is growing in West-East direction.



2. There are two main factors that affecting the bands thickness:

- 1 - The non-constant frequency in time of the magnetic field reversals (main factor); and
- 2 - The rate growth of the Atlantic basin (or new crust formation).

3. To verify the growth rates we need to establish the level of symmetry of the polarity bands on both sides of the ridge. Perfect symmetric polarity bands would indicate both sides grow at the exact same speed.